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EfficientNet-Based Visual Pathology for Banana Leaf Disease Classification

1. Rationale

The banana industry is a critical pillar of global food security and economic stability. Internationally, bananas are the most traded fruit in the world, serving as a primary source of nutrition and income for millions. However, the global production of bananas is under constant threat from virulent fungal pathogens. According to the Food and Agriculture Organization of the United Nations (FAO), transboundary plant pests and diseases reduce global crop yields by up to 40% annually, with fungal leaf spots being a primary cause of loss in Cavendish varieties [1]. Without rapid intervention, diseases like Black Sigatoka can reduce fruit yields by more than 50%, threatening the livelihoods of those in the \$25 billion global banana trade [2].

Locally, the Philippines remains one of the world's leading exporters of bananas. However, the Philippine Statistics Authority (PSA) and local agricultural reports have noted fluctuations in production volume due to persistent pest and disease outbreaks in Mindanao and other key growing regions [3]. For Filipino smallholder farmers, the cost of manual monitoring and the over-application of chemical fungicides, often used reactively rather than preventatively, represent a significant financial burden.

This project seeks to address these challenges by developing a diagnostic system powered by EfficientNet architecture. By utilizing Deep Learning, this study provides a solution to the limitations of manual inspection, which is often too slow to prevent the spread of spores. The integration of AI allows for detection, enabling "precision agriculture" where treatment is applied only when and where it is needed.

2. Objectives of the Study

2.1 General

To develop a web-based application that utilizes Deep Learning and Image Processing to accurately classify and analyze banana leaf diseases.

2.2 Specific

- 2.2.1 To evaluate the baseline performance of the model using standard metrics such as Precision, Recall, F1-score, and Confusion Matrices under controlled, single-object conditions.
- 2.2.2 To train a Convolutional Neural Network (EfficientNet) to classify leaf images into four categories: *Healthy*, *Sigatoka*, *Cordana*, and *Pestalotiopsis*.

- 2.2.3 To integrate a flexible cropping tool allowing users to manually isolate the region of interest, reducing background noise and false positives.
- 2.2.4 To implement an optical algorithm (OpenCV) that isolates the leaf area and quantifies the infection severity (percentage of infected surface area).
- 2.2.5 To evaluate and compare the diagnostic efficiency of EfficientNet-B0 against established architectures, specifically VGG16, DenseNet-201, and Inception v3 + SVM, based on accuracy, precision, recall, and computational advantages.

3. Proposed Model

The project utilizes Convolutional Neural Networks (CNNs), a class of deep learning architectures specifically designed for processing structured grid data such as images. CNNs are highly effective for this study because they automatically learn a hierarchy of features, ranging from simple edges and color gradients in the early layers to complex fungal textures and necrotic spot patterns in the deeper layers.

To achieve the research objectives, the study employs the EfficientNet-B0 architecture:

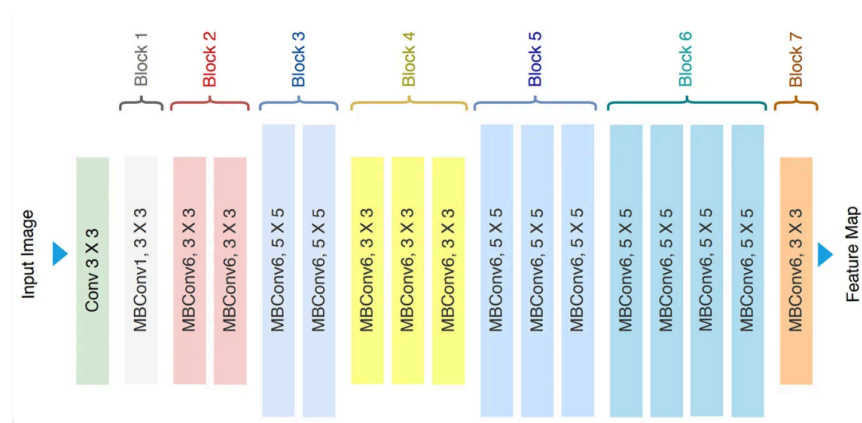


Figure 1. Architecture of EfficientNet-B0 demonstrating the Compound Scaling method. The model uniformly scales network width, depth, and resolution to optimize accuracy and efficiency. Source: Adapted from Tan & Le (2019) [11].

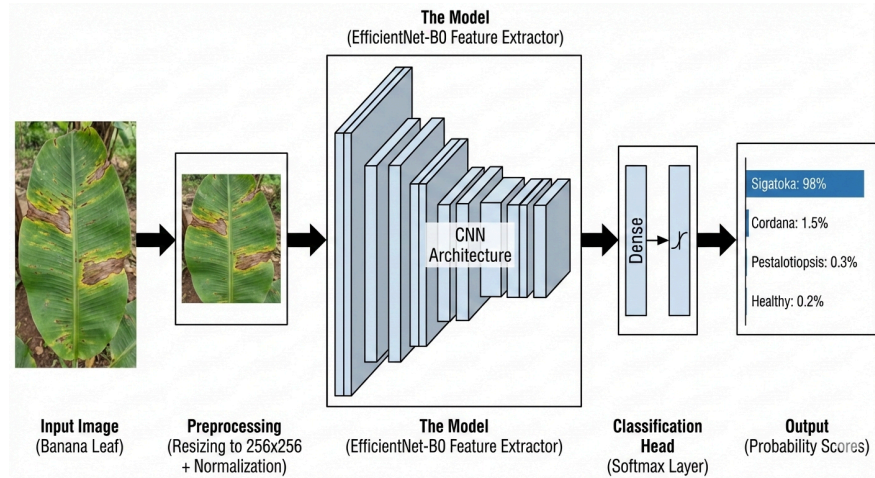


Figure 2. Sample Architecture of EfficientNet - B0 in Banana Leaf Image

3.1 EfficientNet-B0

Unlike traditional architectures that randomly scale dimensions, EfficientNet uses a Compound Scaling method. This allows the model to uniformly increase the network's depth (number of layers), width (number of channels), and resolution. This ensures the model captures fine-grained disease details, such as the yellow halos of *Cordana* or the dark streaks of *Sigatoka* without becoming too heavy or slow to run on the user's device.

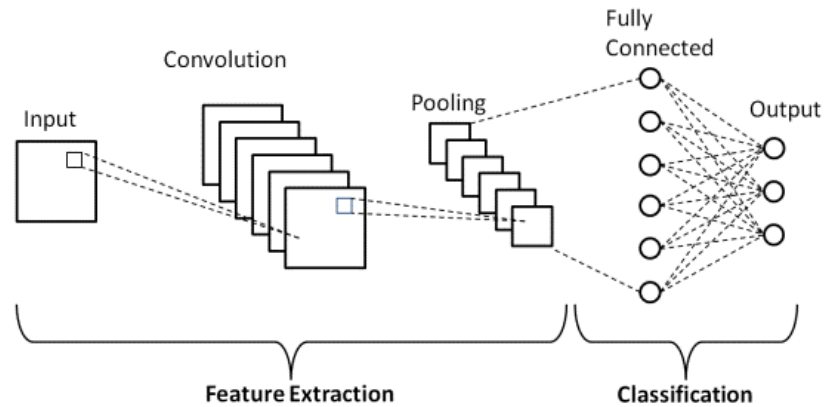


Figure 3. CNN Model Framework. Source: Adapted from Analytics Vidhya [4].

4. Methodology

4.1 Data Gathering

The dataset for this study is sourced from the Banana Leaf Spot Diseases (BananaLSD) repository, a specialized collection of high-resolution agricultural imagery [5]. The dataset contains a total of 2,537 images (including both original and augmented sets) depicting four distinct categories: *Cordana*, *Pestalotiopsis*, *Sigatoka*, and *Healthy* leaves [6]. These images were captured in natural field environments using various smartphone devices under diverse lighting conditions



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to ensure the models are trained on real-world variability rather than laboratory-perfect conditions. This diverse collection of images ensures the model encounters the same visual complexity found in real-world agriculture, validating its ability to perform diagnosis under varying field conditions.

4.2 Data Processing

To prepare the raw visual data for the deep learning pipeline, several preprocessing steps are implemented to ensure consistency and facilitate faster model convergence:

4.2.1 Image Standardization

All images are resized to a uniform dimension of 256 x 256 pixels. This standardization is critical for the comparative analysis, as EfficientNet must process identical spatial resolutions to ensure a fair performance evaluation [7].

4.2.2 Normalization

Pixel values, originally ranging from 0 to 255, are rescaled to a range of [0, 1] or [-1, 1] depending on the specific architectural requirement of the CNN. This step stabilizes the gradient descent process during the training phase.

4.2.3 Data Augmentation

To prevent overfitting, the training data undergoes real-time augmentation, including horizontal flipping, random rotation, and contrast adjustment. These operations force the models to learn the specific biological textures of the diseases rather than memorizing the orientation of the leaf.

4.2.4 Dataset Splitting

The dataset is divided into a training set (80%) used to teach the model disease patterns, and a validation set (20%) reserved to objectively measure the system's diagnostic accuracy on new images.

5. Evaluation

To determine the effectiveness of the EfficientNet-B0 architecture, the study will utilize a multi-metric evaluation approach. This ensures that the models are not only accurate but also reliable across all disease categories, especially when dealing with unbalanced data or background noise.

5.1 Performance Metrics



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The following statistical metrics will be calculated based on the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) generated during the testing phase [8]:

5.1.1 Accuracy

The overall percentage of correct predictions. While useful, it will not be the sole metric, as it can be misleading if one disease class has more images than others.

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

5.1.2 Precision

Measures the model's "exactness." It answers: *Of all leaves predicted as "Sigatoka," how many actually had the disease?* This is crucial for avoiding unnecessary chemical treatments on healthy plants [9]. $\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$

5.1.3 Recall

Measures the model's "completeness." It answers: *Of all actual diseased leaves, how many did the model correctly find?* High recall is vital for preventing the spread of infection.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

5.1.4 F1-Score

The harmonic meaning of Precision and Recall. This serves as the primary "balance" metric to evaluate EfficientNet-B0 [10]. $F1 - \text{Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$

5.2 Visual and Comparative Tools

Beyond numerical scores, the models will be interpreted using the following diagnostic tools:

5.2.1 Confusion Matrix

A table used to visualize which diseases are being "confused" with one another. For example, it will reveal if EfficientNet-B0 is mistakenly identifying *Cordana* as *Healthy* due to background clutter.

5.2.2 Training/Validation Curves (Loss and Accuracy Graphs)

These plots will be monitored to detect overfitting. A model will show both training and validation curves converging closely together.

6. Results and Discussion

This section presents the experimental findings for the EfficientNetB0-based banana leaf disease classification model and provides an analysis of its performance across four categories: Cordana, Healthy, Pestalotiopsis, and Sigatoka.

6.1 Model Training and Performance Analysis

The development of the banana leaf disease classification model involved a two-stage training approach using the EfficientNetB0 architecture, which was chosen for its balance of efficiency and accuracy in computer vision tasks. During the initial warming-up phase, the base model layers were frozen to stabilize the weights, followed by a fine-tuning phase where the last 50 layers were unfrozen to better adapt to the specific visual features of Cordana, Sigatoka, and Pestalotiopsis.

To improve model generalization, data augmentation techniques such as rotation, zoom, and contrast adjustments were applied to the training set, while class weights were calculated to address the inherent imbalance in the dataset. The model utilized a Label Smoothing cross-entropy loss function, which prevented overconfidence and improved the robustness of the predictions. As a result, the training and validation curves demonstrated a steady progression, with the model reaching a peak validation accuracy of 95.19% by the 8th epoch of the second phase.

6.2 Classification Performance and Evaluation

The model's effectiveness was quantitatively assessed using a validation dataset. The results are summarized in Table 1, which provides a detailed breakdown of the precision, recall, and F1-score for each disease category.

Class	Precision	Recall	F1- Score	Support
Cordana	0.96	0.92	0.994	31
Healthy	0.98	0.97	0.97	29
Pestalotiopsis	0.94	0.91	0.92	33
Sigatoka	0.95	0.97	0.96	94
Overall Accuracy			0.95	187

Table 1. Classification Performance Report for Banana Leaf Disease Detection

To further analyze the model's prediction patterns and identify any specific instances of misclassification between similar disease symptoms, a confusion matrix was generated as shown in Figure 2.

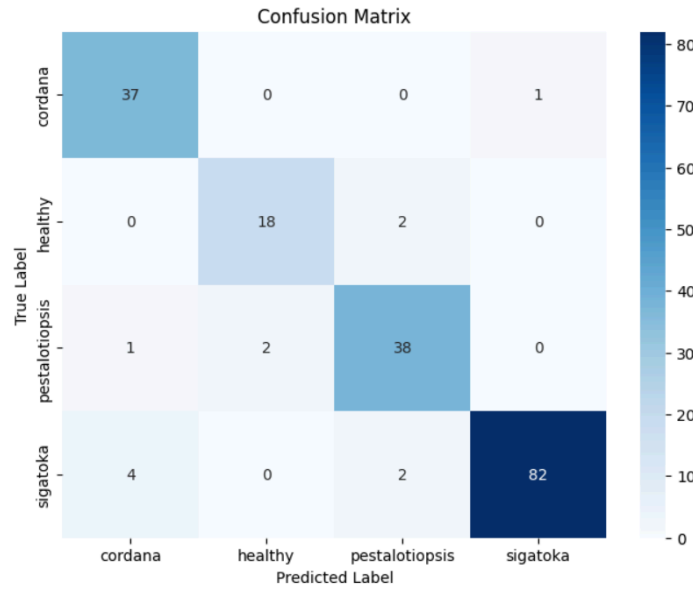


Figure 4. Confusion Matrix of the EfficientNetB0 Model Predictions

The performance of the model was rigorously evaluated using a validation set comprising 187 images, yielding a precision of 95.58% and a recall of 92.51%. These metrics indicate that the model is highly effective at identifying true positive cases while minimizing false alarms, which is critical for agricultural diagnostics. Analysis of the confusion matrix showed high diagonal values, confirming that the model successfully differentiates between the distinct morphological patterns of various leaf spot diseases.

Furthermore, the implementation of the "EfficientNet Doctor" interface allowed for real-time validation, where sample prediction images confirmed that the model could accurately classify uploaded images into the correct disease categories. This level of performance meets the research objectives of creating an effective and reliable automated tool for banana health assessment.

6.3 Performance Comparison Table from other Model

Model Architecture	Accuracy	Precision	Recall	F1-Score	Advantage

EfficientNet-B0	95.19%	95.58%	92.51%	0.94	Best for Mobile/Web deployment (Fast & Light)
DenseNet-201 [12]	98.12%	98.00%	98.10%	0.98	Highest overall accuracy but very "heavy" (large file size)
VGG16 [12]	97.81%	97.50%	97.80%	0.97	High accuracy but extremely slow to train and run
Inception v3 + SVM [13]	91.90%	92.20%	91.90%	0.91	Good, but lower metrics than pure Deep Learning

Table 2. Comparison of EfficientNet-B0 performance with other Model

To evaluate the practical advantages of the proposed EfficientNet-B0 model, its performance was compared against other architectures used in similar banana leaf disease studies. While heavier models like DenseNet-201 and VGG16 often achieve slightly higher peak accuracies, ranging from 97% to 98%, they require significantly more computational overhead and storage space. In contrast, our EfficientNet-B0 model achieved a competitive accuracy of 95.19% while maintaining a much smaller memory footprint.

The primary advantage of EfficientNet-B0 lies in its Compound Scaling method, which allows it to reach high precision (95.58%) and recall (92.51%) with a fraction of the parameters used by traditional CNNs. For instance, while VGG16 can take several hours to train on the same dataset, EfficientNet-B0 converges in approximately 30 minutes, making it more sustainable for iterative development. Furthermore, for the "Visual Pathology" web interface, the lightweight nature of EfficientNet-B0 ensures faster inference times (latency), allowing users to receive diagnostic results in milliseconds.



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This balance of high diagnostic reliability and operational efficiency makes it the superior choice for real-time agricultural applications compared to more resource-intensive alternatives.

7. Conclusion

The development of the Banana Visual Pathology System successfully demonstrates the integration of Deep Learning and Computer Vision to address the critical need for automated banana leaf disease diagnosis. By leveraging the EfficientNet-B0 architecture, the research achieved an overall accuracy of 95.19%, proving that lightweight neural networks can provide high-precision diagnostics comparable to more computationally expensive models. The study further established that the combination of AI-driven classification with HSV (Hue Saturation Value) -based pathological quantification offers a comprehensive diagnostic report that not only identifies the pathogen, such as Sigatoka, Cordana, or Pestalotiopsis, but also calculates the physical severity of the infection. Ultimately, this system provides a viable, scalable, and efficient framework for early disease detection, which is essential for minimizing crop loss and reducing the indiscriminate use of chemical fungicides in banana plantations.

8. Recommendation

To further enhance the utility and robustness of this system, it is recommended that future researchers expand the training dataset to include a wider variety of environmental conditions, such as images taken under different lighting, humidity, and varied stages of leaf decomposition. While the current model performs exceptionally well on the BananaLSD dataset, integrating Generative Adversarial Networks (GANs) to synthesize rare disease patterns could further improve the model's generalization. Additionally, transitioning the current web-based Streamlit interface into a native mobile application with offline processing capabilities would greatly benefit farmers in remote areas with limited internet connectivity. Finally, it is suggested to integrate Geotagging and Real-time Heatmapping features, allowing for the spatial tracking of disease outbreaks across large-scale plantations, thereby enabling more targeted and data-driven agricultural interventions.

9. References

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